

Retail Product Performance, Unit Economics & Price Elasticity

Daily trading analysis for a single retail product across 731 trading days (01 Jan 2023 – 31 Dec 2024). All financial metrics are denominated in South African Rand (ZAR). This report details unit economics, margin invariants, temporal demand dynamics, price elasticity of demand (PED), and promotional strategy recommendations.

TOTAL SALES

R 20 391 531

731 trading days

TOTAL GROSS PROFIT

R 5 118 621

Cumulative gross margin

GROSS PROFIT MARGIN

25.1%

Blended overall margin

UNITS SOLD

111 623

Total volume moved

1. Unit Economics

TOTAL SALES PRICE / UNIT

R 182,68

$R\ 20\ 391\ 531 \div 111\ 623$
units

AVG UNIT SALES PRICE

R 182,68

Avg unit cost: R 136,83

AVG GROSS PROFIT / UNIT

R 45,86

25.1% profit per unit

AVERAGE DAILY SALES

R 27 895

153 units moved per day

Daily Sales Price per Unit

Sales $\div$ quantity sold per trading day. Highlighted yellow dips represent promotional campaigns.



Daily % GP vs. % GP per Unit

Both series are identical by construction — margin is scale-invariant, validating unit calculations.



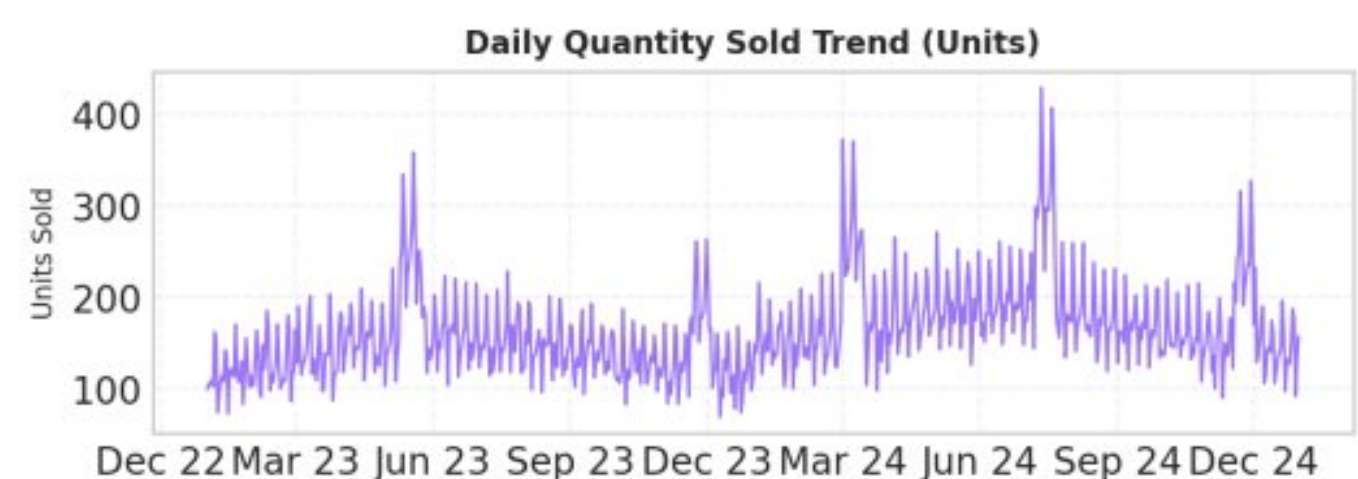
Daily Sales Trend

Total Rand value of sales per trading day across 2023–2024.



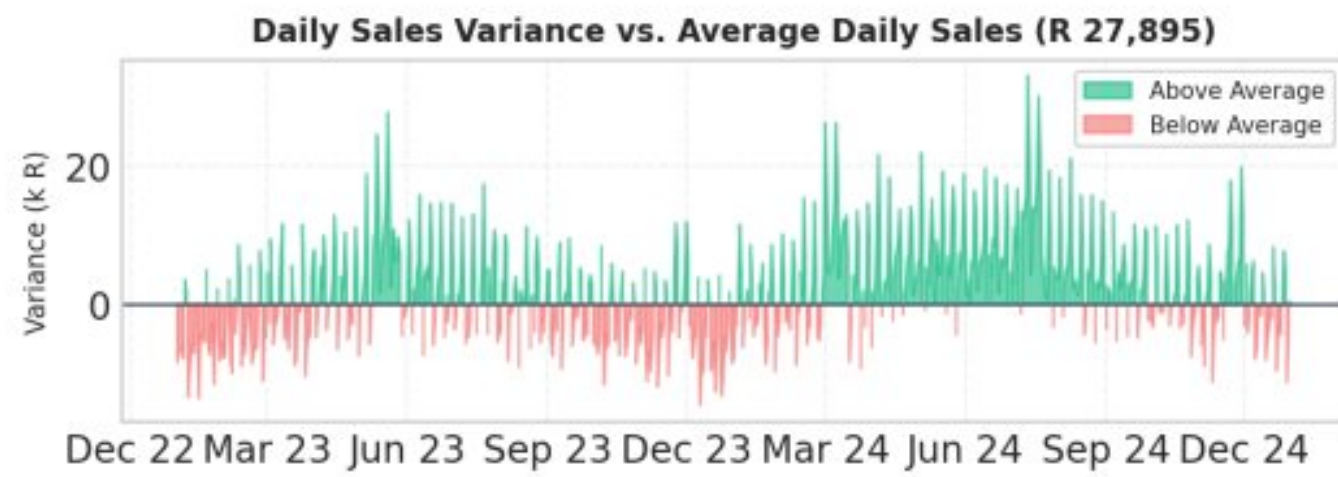
Daily Quantity Sold Trend

Units moved per day, displaying sharp surges during discounted promotional trading windows.



Daily Sales vs. Average Daily Sales

Daily variance against the R 27 895 baseline (green = above average, red = below average).



Cumulative Sales Growth

Running total of Rand sales accumulating to R 20.39 Million by year-end 2024.



2. Monthly & Annual Performance

Monthly Sales vs Cost of Sales

Rand value comparison of revenue versus inventory cost per calendar month.



Total Profit by Month

Gross profit in Rands (bars) alongside gross profit margin percentage overlay (line).



Total Quantity Sold by Month

Volume moved per calendar month across the 24-month trading span.



Gross Profit % Distribution by Month

Daily margin range (min - max grey bar) with the monthly median (blue markers).



Profit by Year

Sales, Cost of Sales, and Gross Profit comparison (2023 vs 2024).



2023 GROSS PROFIT

R 3 180 916

R 9 543 093 sales · 33.3% margin · 49 996 units

2024 GROSS PROFIT

R 1 937 705

R 10 848 438 sales · 17.9% margin · 61 627 units

YEAR-ON-YEAR SALES GROWTH

+13.7%

Volume +23.3% (61 627 vs 49 996 units), but GP declined -39.1%

3. Demand Behaviour & Price Elasticity

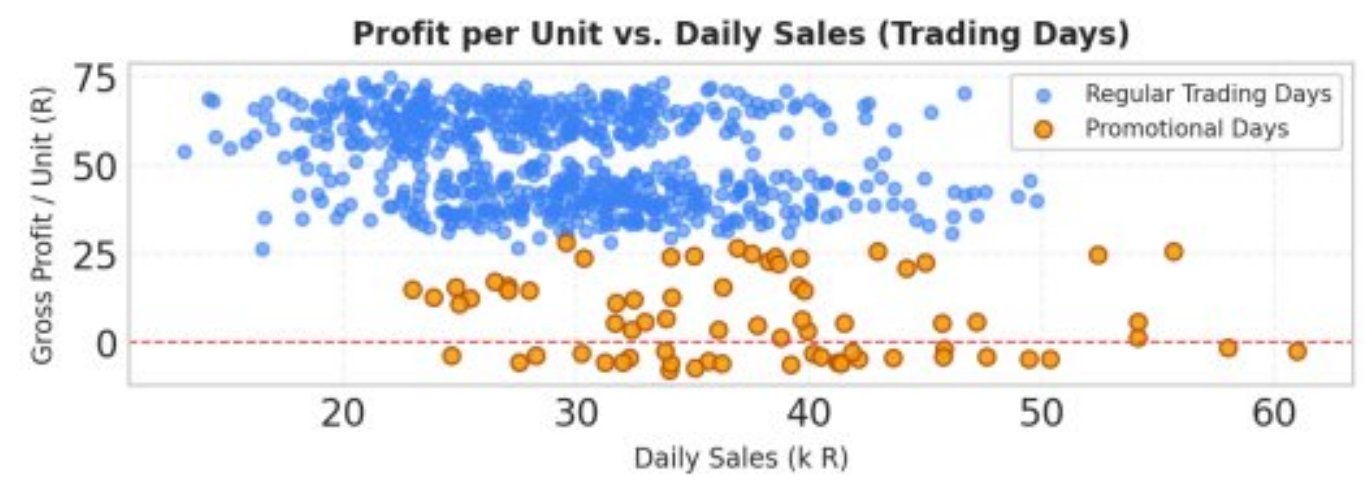
Average Sales by Day of Week

Mean daily Rand sales (bars) and units sold (line) by weekday.



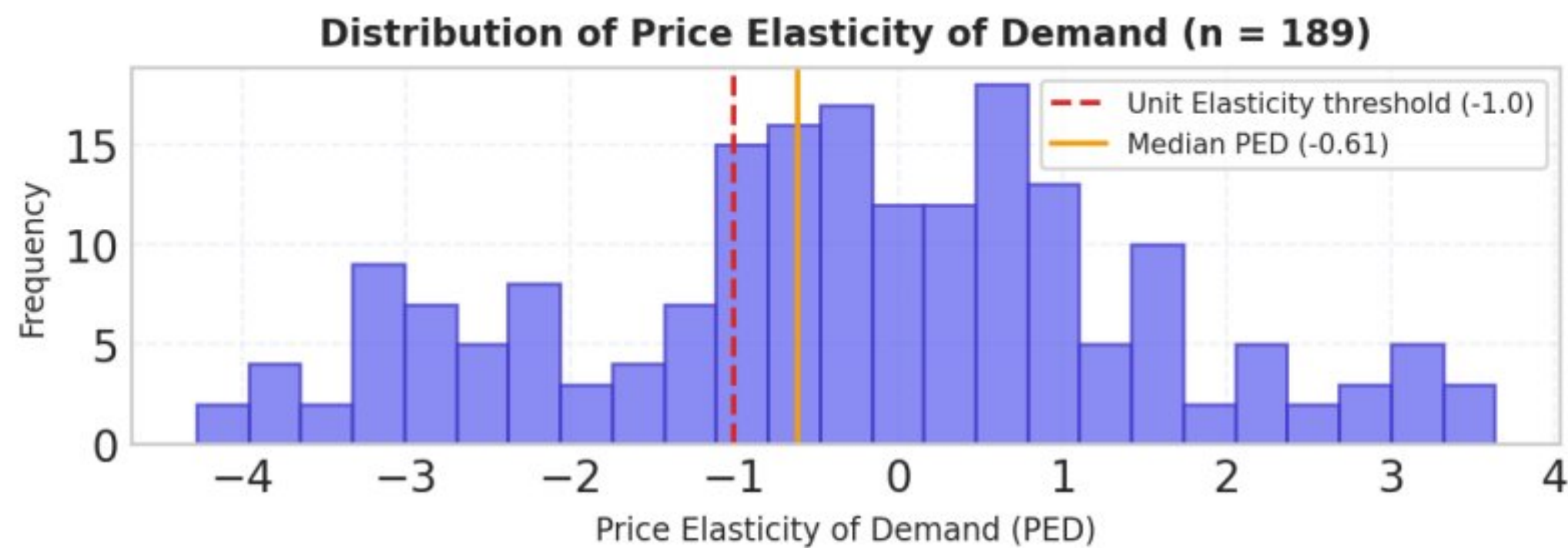
Profit per Unit vs. Daily Sales

Each point represents one trading day — promotional days highlighted in amber.



Distribution of Price Elasticity of Demand

Day-on-day point elasticities (n = 189); values left of -1 indicate elastic demand. Mean elasticity 0.02 · median -0.61 · 48% of price moves produced elastic demand.



Promotional Periods — Price Elasticity of Demand (PED)

Promo window vs. the 14 trading days either side ($PED = \% \Delta \text{ quantity} \div \% \Delta \text{ price}$)

PERIOD	PROMO PRICE	%Δ PRICE	%Δ QUANTITY	PED	DAILY GROSS PROFIT
Promo 1 10 May 23 – 24 May 23	R 155,08 vs R 198,38	-21.8%	+58.9%	-2.70	R 6 948 -35.0% vs normal
Promo 2 20 Nov 23 – 04 Dec 23	R 151,13 vs R 193,64	-22.0%	+54.9%	-2.50	R 3 314 -55.7% vs normal
Promo 3 01 Mar 24 – 14 Mar 24	R 145,35 vs R 185,71	-21.7%	+70.5%	-3.24	R 1 606 -76.6% vs normal
Promo 4 08 Jul 24 – 21 Jul 24	R 140,49 vs R 180,71	-22.3%	+79.5%	-3.57	-R 1 502 -123.7% vs normal
Promo 5 18 Nov 24 – 02 Dec 24	R 144,59 vs R 185,60	-22.1%	+54.8%	-2.48	-R 1 645 -133.1% vs normal

4. Findings & Recommendations

1. Demand is Strongly Price Elastic

Evidence: Average promotional PED is **-2.90**, meaning each 1% reduction in retail price yields approximately a 2.9% surge in unit volume. Promotional campaigns elevated daily revenue by **27.8%**, while daily gross profit fell by **-84.8%** due to unhedged unit cost escalation.

Recommendation: Keep running the ~20% discount mechanic strategically — it is highly volume-accretive. However, discounts deeper than 25% must be strictly pre-tested, as severe unit cost inflation in 2024 resulted in negative gross profits during Promos 4 & 5.

2. Trade to the Weekend Peak

Evidence: Friday trading generates mean sales of **R 37 494** compared to Sunday at **R 21 848** — representing a 72% weekend surge gap. In terms of monthly volume, July 2024 represented the peak sales window at **R 1 169 022**.

Recommendation: Synchronize inventory replenishment schedules, staffing allocations, and promotional launch days to run Thursday through Saturday. Secure buffer stock ahead of the high-velocity November–December year-end peak.

3. Protect the Margin Floor

Evidence: While the 2-year blended gross margin stands at **25.1%**, underlying unit cost drifted upward throughout 2024 without matching adjustments in regular shelf prices. Consequently, 2024 gross profit contracted to R 1.94M (17.9% margin) despite 13.7% revenue expansion.

Recommendation: Establish formal quarterly list-price reviews benchmarked against landed supplier costs. Implement a hard promotional margin floor guardrail ensuring promotional unit price never breaches the monthly median cost baseline.

Dataset Footnote: This analysis is powered by the 731-day daily trading dataset conforming to standard retail analytics schemas (Date, Sales, Cost of Sales, Quantity Sold) denominated in South African Rand (ZAR).